

Case Study: Creative Developer (Freelance)

Project: Number Match

About this case study

Thanks for taking the time to work through this. Below is a short case study built around Number Match, a small puzzle mini-game. You'll be picking up where we've left off and building out the core interaction.

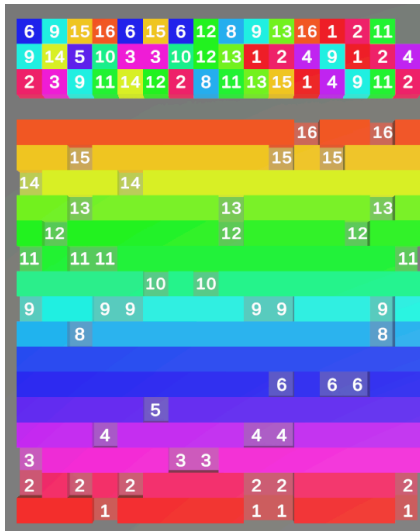
You'll be given a working Unity project with the game already generating its board and pieces. Your job is to implement the interactive gameplay layer and the reveal feel on top of it.

The project: Number Match

Number Match is a small “reveal the picture” puzzle. A grid is generated from a source image, where each cell is assigned the number of its closest matching color. Some cells start already showing their true color, while the rest start masked, showing only their number. Loose numbered cubes are laid out above the board. The player's job is to pick up each cube and place it on a grid cell with the same number. Once every cell belonging to a number has been filled in, that number's color is revealed across the whole board, so the full picture gradually appears.

What you're given

A Unity 6 project where the board and tray already generate correctly. The grid is built from a source image, and the leftover numbered cubes are laid out above it. None of the interaction or reveal logic exists yet, and that's what we're asking you to build.



What we'd like you to add

Task	Requirement
1	Mask the full painting with a flat neutral gray. No color should be visible anywhere on the board until a piece has been correctly placed and its number group is complete.
2	Let the player pick up a numbered cube from the tray and place it onto the matching numbered slot on the grid (drag and drop). Wrong placements should give clear negative feedback and return the cube to the tray.
3	When every slot belonging to a given number has been filled, reveal that number's true color across all of its cells.
4	Add a satisfying reveal effect for the moment a number group completes, such as the gray mask breaking, shattering, or dissolving away to show the real color underneath. This is the part we care about most, so push it further than the minimum if you can: sequencing, easing, particles, camera feel, and audio hooks are all fair game. Style is your call.
5	Keep it running smoothly on a larger grid. The current setup spawns one object per cell, so a bigger board gets expensive quickly. Show us how you'd keep the frame rate healthy while the reveal effects are firing.

Development time for this case study is 3 days. If you finish sooner, feel free to send it in early.

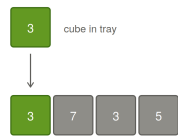
What each task looks like

Task 1: mask the painting gray



no color visible, numbers only

Task 2: place a numbered cube



wrong number bounces back to tray

Task 3: reveal the number's color



all 3s filled, so every 3 turns color

Task 4: satisfying reveal effect



gray mask breaks away, color underneath

What matters most to us

This is a creative developer role, so two things carry the most weight. First, how the reveal looks and feels: we want effects that are genuinely appealing to watch, not just functional. Second, how it holds up at scale: the reveal should still feel smooth when the grid gets large and many cells are animating at once.

Everything else in the list is the baseline we need in place so those two things can be judged fairly.

Bonus

If you have time and an idea you like, add anything you want on top: an extra mechanic, an effect, a feature, whatever you think would make the game better. This is completely open and entirely optional, but it's a good chance to show us what you'd bring to a project beyond the brief.

A few notes to save you time

- Build and tune on a small grid if that's easier, but make sure it also holds up on a larger one. Tell us what size you tested at.

- Feel free to structure the placement and reveal logic however makes sense to you. We're more interested in how you think about it than in matching any particular pattern.
- The illustrations above are rough sketches to show intent. They're not a visual spec, so don't feel tied to them.

How to send it back

- The Unity project with a link to a repo or zip.
- A short screen recording of the final result.
- A note that describes your approach, the grid size you tested at, and any assumptions you made.